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THE FUNCTIONAL EQUATION $f \circ f + af + b1_{\mathbb{R}} = 0$

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Abstract. We compute all the continuous solutions of the functional equation in the title, for real a, b and $b \neq 0$. The solutions are obtained using different patterns which depend upon the solutions of the associated characteristic equations.

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0. Preliminaries

Throughout the paper $\mathbb{R}$ will be the real field, $\overline{\mathbb{R}}$ will be the extended real field, $\mathbb{N}$ will be the set of natural numbers and $\mathbb{Z}$ will be the set of integer numbers.

If $A \subset \mathbb{R}$ is a non empty set and $f : A \to \mathbb{R}$, we shall write $f \uparrow$ or $f(x) \uparrow$ (respectively $f \downarrow$ or $f(x) \downarrow$) to denote the fact that f is increasing (respectively decreasing). If the monotony is strict, we shall write $f \uparrow$ (strictly) or $f(x) \uparrow$ (strictly) (respectively $f \downarrow$ (strictly) or $f(x) \downarrow$ (strictly)) For sequences we use the same symbols: if $(x_n)_n$ is a sequence of real numbers we shall write $x_n \uparrow$ and $x_n \uparrow$ (strictly) (respectively $x_n \downarrow$ and $x_n \downarrow$ (strictly)) to denote the fact that $(x_n)_n$ is increasing or strictly increasing (respectively decreasing or strictly decreasing). If the limit of such a sequence is specified, we shall write $x_n \uparrow l$ or $x_n \uparrow l$ (strictly) (respectively $x_n \downarrow l$ and $x_n \downarrow l$ (strictly)) where $l \in \overline{\mathbb{R}}$. More general, we write $x_n \underset{n}{\to} l$ to denote the fact that $l \in \overline{\mathbb{R}}$ is the limit of $(x_n)_n$.

Let $A \subset \mathbb{R}$ be a non empty set and $f : A \to A$ be a function. The identity of A is the function $1_A : A \to A$, acting via $1_A(x) = x$ for any

$x \in A$. We can compute the compositions $f \circ f = f^2$ and f^n for any natural n (with the conventions $f^0 = 1_A$ and $f^1 = f$). If f is bijective, we can compute f^{-1}, $f^{-2} = f^{-1} \circ f^{-1}$ and f^{-n} for any natural $n \geq 1$.

A fixed point of f is a point $x \in A$ such that $f(x) = x$. We shall write $fix(f) = \{x \in A \mid x \text{ is a fixed point of } f\}$.

Finally, let us us recall the fact that, in case I and J are non degenerate intervals and $h : I \to J$ the following assertions are equivalent:

a) h is a homeomorphism;

b) h is a continuous bijection;

c) h is a (strictly) monotone bijection;

d) h is a strictly monotone surjection.

For the general theory of functional equations, see [2] and [3]. Interferences between the present paper and [1] appear in the particular case $a = 0$ and $g(x) = -bx$.

The authors wish to lay stress upon the following facts. After writing our paper we found out the existence of the paper [4], which is concerned with the same subject. Because the majority of our proofs differ from the proofs in [4], we consider that the present paper can be of some interest.

1. Solving the functional equation: the beginning

Let a, b be real numbers, $a \neq 0$, $b \neq 0$. We shall be concerned with the functional equation (called *fundamental equation*)

$$f \circ f + af + b1_{\mathbb{R}} = 0.$$

Namely, we want to find a continuous function $f : \mathbb{R} \to \mathbb{R}$ having the property that, for any $x \in \mathbb{R}$

$$f(f(x)) + af(x) + bx = 0. \tag{1.0}$$

Such a function (in case it exists) will be called *a solution of the fundamental equation* (or, simply *a solution*). In the sequel, the fundamental equation will be written in the form

$$f \circ f + af + bx = 0. \tag{1.1}$$

Incidentally, the fundamental equation (1.1) will be written alternatively

$$f \circ f \pm af \pm bx = 0. \tag{1.1'}$$

with *positive* a and b.

The *characteristic equation* of the problem is the equation

$$x^2 + ax + b = 0 \tag{1.2}$$

with (complex) roots r_1, r_2 and discriminant $\Delta = a^2 - 4b$.

In case r_1 and r_2 are real, one sees immediately that the functions $S(r_i) : \mathbb{R} \to \mathbb{R}$, given via $S(r_i)(x) = r_i x, \quad i = 1, 2$ are solutions.

In case $r_1 = 1$ (respectively $r_2 = 1$) one can see that $S(r_2) + C$ (respectively $S(r_1) + C$) for any $C \in \mathbb{R}$ is a solution. In case $r_1 \neq 1$ and $r_2 \neq 1$ (r_i being real), a function of the form $S(r_i) + C$ is a solution if and only if $C = 0$. Moreover, in this case, a solution can be equal to $S(r_i) + C$ on a non degenerate interval only if $C = 0$.

Theorem 1.1. *A solution f is a homeomorphism.*

Proof. Let $f : \mathbb{R} \to \mathbb{R}$ be a solution of (1.1).

a) The function f is *injective*. Indeed, assume by absurd the existence of $x_1 \neq x_2$ such that $f(x_1) = f(x_2)$. Hence $f(f(x_1)) + af(x_1) + bx_1 = f(f(x_2)) + af(x_2) + bx_2 = 0$ which is not possible because $b \neq 0$.

Due to the continuity of f it follows that f must be strictly monotone.

b) The function f is *surjective*. Indeed, due to the monotony of f, there exist $\lim_{x \to -\infty} f(x) = l \in \overline{\mathbb{R}}$ and $\lim_{x \to \infty} f(x) = L \in \overline{\mathbb{R}}$.

Assume, e.g. that f is increasing. We prove that $L = \infty$. If not, let $L \in \mathbb{R}$. Taking an increasing sequence $x_n \underset{n}{\to} \infty$ we have $f(x_n) \to L$, $f(f(x_n)) \underset{n}{\to} f(L)$, hence $f(f(x_n)) + af(x_n) + bx_n \underset{n}{\to} \pm\infty$ (according to $b > 0$ or < 0) and this contradicts (1.0). In the same way, one sees that $l = -\infty$ and this proves the surjectivity of f.

The bijective functions f and f^{-1} are monotone, hence they are continuous. □

Let us notice the importance of the condition $b \neq 0$ which guarantees the fact that constant functions cannot be solutions (în case $b = 0$ the null function is a solution whereas in case $b = 0$ and $a = -1$ every constant function is a solution).

In case $a = 0$ and $b \neq 0$, we have solutions only in case $b < 0$, because $f \circ f \uparrow$. This case will appear as a particular case of [1]. Notice that the case $a = 0$ and $b = -1$ is a particular case of the Babbage equation $f^n = x$ (see [3]).

Before passing further, let us use Theorem 2.1. and take $x = f^{-2}(t)$ in (1.0). We get

$$t + af^{-1}(t) + bf^{-2}(t) = 0 \Leftrightarrow f^{-2}(t) + \frac{a}{b}f^{-1}(t) + \frac{1}{b}t = 0 \tag{1.3}$$

which is the functional equation fulfilled by f^{-1} (with possible variations, see (1.1')).

Consequently, the characteristic equation of problem (1.3) is

$$x^2 + \frac{a}{b}x + \frac{1}{b} = 0 \tag{1.4}$$

and has the roots $\frac{1}{r_1}$, $\frac{1}{r_2}$. Equations (1.3) and (1.4) show that the problems of finding a solution f or its inverse f^{-1} are equivalent. This fact will be used often in the sequel. For example, the case $1 < r_1 < r_2$ (i.e. $0 < \frac{1}{r_2} < \frac{1}{r_1} < 1$) is equivalent to the case $0 < r_1 < r_2 < 1$. We shall see in the sequel how to construct the solutions in case $1 < r_1 < r_2$. Then, we shall be able to construct the solutions in case $0 < r_1 < r_2 < 1$. Namely, we have $1 < \frac{1}{r_2} < \frac{1}{r_1}$. Firstly we construct the solutions for the characteristic equation having $\frac{1}{r_1}$ and $\frac{1}{r_2}$ as roots and let S be the set of these solutions. Then, the set of solutions for the initial problem will be $\left\{f^{-1} \mid f \in S\right\}$. We can use also the obvious facts that $S(r)^{-1} = S\left(\frac{1}{r}\right)$ and $\left(S(r) + c\right)^{-1} = S\left(\frac{1}{r}\right) - \frac{c}{r}$.

Now, we shall see that the existence of real roots for the characteristic equation is a necessary and sufficient condition for the existence of solutions.

Theorem 1.2 (Case $\Delta < 0$)**.** *Assume $\Delta < 0$. Then the equation* (1.1) *has no solutions.*

Proof. Assume f is a solution of (1.1). We shall arrive at a contradiction. Clearly, $b > 0$.

From the equation $x + af^{-1}(x) + bf^{-2}(x) = 0$ we obtain, taking $x = f^{-n+2}(y)$ with n natural $f^{-n+2}(y) + af^{-n+1}(y) + bf^{-n}(y) = 0$, hence, for $y < x$ and $n \geq 2$

$$\begin{aligned}&\left(f^{-n+2}(x) - f^{-n+2}(y)\right) + a\left(f^{-n+1}(x) - f^{-n+1}(y)\right)\\ &+ b\left(f^{-n}(x) - f^{-n}(y)\right) = 0\end{aligned}$$

and, finally

$$\frac{f^{-n+2}(x) - f^{-n+2}(y)}{f^{-n+1}(x) - f^{-n+1}(y)} + a + b\frac{f^{-n}(x) - f^{-n}(y)}{f^{-n+1}(x) - f^{-n+1}(y)} = 0.$$

Writing

$$y_n = \frac{f^{-n+1}(x) - f^{-n+1}(y)}{f^{-n}(x) - f^{-n}(y)}$$

we have

$$y_{n-1} + a + \frac{b}{y_n} = 0 \Rightarrow y_n = -\frac{b}{a + y_{n-1}}. \tag{1.5}$$

In case $a > 0$, we have $f \downarrow$ because $-af = f \circ f + bx \uparrow$ and $y_n < 0$, hence $a + y_{n-1} > 0$ and $y_{n-1} > -a$. It follow that the sequence $(y_n)_n$ is bounded. At the same time $y_n = u(y_{n-1})$, where $u : (-a, 0) \to \mathbb{R}$, $u(x) = -\frac{b}{a+x}$ is strictly increasing, therefore $(y_n)_n$ is monotone and convergent.

In case $a < 0$, we have $f \uparrow$ because $-af = f \circ f + bx \uparrow$ and $y_n > 0$, hence $a + y_{n-1} < 0$ and $y_{n-1} < -a$. It follow that $(y_n)_n$ is bounded. Because $y_n = r(y_{n-1})$ where $r : [0, -a) \to \mathbb{R}$, $r(x) = -\frac{b}{a+x}$ is strictly increasing, it follows that $(y_n)_n$ is monotone and convergent,

In all cases, let $l = \lim_n y_n$. From(1.5) we get $l = -\frac{b}{a+l} \Leftrightarrow l^2 + al + b = 0$ which is impossible, because $\Delta < 0$. $\square$

Theorem 1.3 (Case $r_2 < 0 < r_1$)**.** *In case $r_1 \neq 1$, the solutions are $S(r_1)$ and $S(r_2)$. In case $r_1 = 1$, the solutions are $S(1)$ and $S(r_2)+c$, $c \in \mathbb{R}$.*

Proof. We must prove that any solution f is of the form in the enunciation. Because $r_1 r_2 < 0$, we shall write $b = -r_1 r_2 > 0$. The fundamental equation is

$$f \circ f + af - bx = 0. \tag{1.6}$$

We can assume $a > 0$, because, in case $a < 0$, the fundamental equation for f^{-1} is $g \circ g - \frac{a}{b} g - \frac{1}{b} x = 0$ with $-\frac{a}{b} > 0$ and f^{-1} is in the same situation as f.

In the same way, we can assume $a < 0$.

A. Let us assume $a > 0$ and try to find the increasing solutions ($f = S(r_1)$ is such a solution).

To start, we write (1.6) in the form

$$f \circ f = bx - af. \tag{1.7}$$

The idea is to find two real sequences $(x_n)_n$, $(y_n)_n$ such that , for any $n \geq 1$, the function $F_n : \mathbb{R} \to \mathbb{R}$, given via

$$F_n(x) = (-1)^{n+1} x_n x + (-1)^n y_n f(x) \uparrow . \tag{1.8}$$

Using (1.7), we start with $x_1 = b$, $y_1 = a$.

Accept (1.8). Because $f \uparrow$, if follows that $F_n \circ f \uparrow$ i.e. $(-1)^{n+1} x_n f(x) + (-1)^n y_n f \circ f(x) \uparrow$ which means (see (1.7)) that $F_{n+1}(x)=(-1)^n y_n bx + (-1)^{n+1} (x_n + ay_n) f(x) \uparrow$ i.e. $F_{n+1}(x)=(-1)^{n+2} x_{n+1} x+(-1)^{n+1} y_{n+1} f(x) \uparrow$ with $x_{n+1} = by_n$, $y_{n+1} = x_n + ay_n$ and all x_n and y_n are strictly positive.

We have $y_{n+2} = x_{n+1} + ay_{n+1} = ay_{n+1} + by_n$ hence

$$\begin{aligned} & y_n = c_1(-r_1)^n + c_2(-r_2)^n \\ & x_n = bc_1(-r_1)^{n-1} + bc_2(-r_2)^{n-1} = c_1 r_2 (-r_1)^n + c_2 r_1 (-r_2)^n. \end{aligned}$$

We have $r_1 + r_2 = -a < 0$, hence $|r_1| < |r_2|$ and it follows that

$$\lim_n \frac{x_n}{y_n} = r_1. \tag{1.9}$$

From (1.7): $-x_{2k} x + y_{2k} f(x) \uparrow$; $x_{2k+1} x - y_{2k+1} f(x) \downarrow$ and, dividing by $x_n > 0$, $y_n > 0$:

$$f(x) - \frac{x_{2k}}{y_{2k}} f(x) \uparrow; \qquad f(x) - \frac{x_{2k+1}}{y_{2k+1}} x \uparrow .$$

By passing to $k-$limit, one obtains (see (1.9)) that $f(x) - r_1 x \uparrow$ and $f(x) - r_1 x \downarrow \Leftrightarrow f(x) - r_1 x = c$ (constant). Because $r_2 \neq 1$, we known that the constant $c = 0$ and $f(x) = r_1 x$, so $f = S(r_1)$.

B. Let us find the *decreasing solutions* ($S(r_2)$ is such a solution). The idea is now to find two real sequences $(x_n)_n$, $(y_n)_n$ such that the function $G_n : \mathbb{R} \to \mathbb{R}$, given via

$$G_n(x) = (-1)^n x_n x + (-1)^n y_n f(x) \downarrow \tag{1.10}$$

under the assumption $a < 0$.

To start, we write the equation in the form $f \circ f - af - bx = 0$ with $a > 0$, $b > 0$. Hence $f \circ f = af + bx \uparrow \Rightarrow -af - bx \downarrow$ and we take $x_1 = b$, $y_1 = a$.

Assuming (1.10), we obtain that $G_n \circ f \uparrow$ (because $f \downarrow$), hence

$$\begin{aligned} G_{n+1}(x) &= (-1)^n x_n f(x) + (-1)^n y_n f \circ f(x) \\ &= (-1)^n x_n f(x) + (-1)^n y_n (af(x) + bx) \uparrow \end{aligned}$$

which means

$$\begin{aligned} G_{n+1}(x) &= (-1)^{n+1} x_{n+1} x + (-1)^{n+1} y_{n+1} f(x) \\ &= (-1)^{n+1} y_n bx + (-1)^{n+1}(ay_n + x_n) f(x) \downarrow . \end{aligned}$$

Hence $x_{n+1} = by_n, \qquad y_{n+1} = x_n + ay_n$ and all x_n, y_n are strictly positive. We obtain $y_{n+2} = ay_{n+1} + by_n$ hence

$$\begin{aligned} y_n &= c_1 r_1^n + c_2 r_2^n \\ x_n &= by_{n-1} = -r_1 r_2 y_{n-1} = -c_1 r_2 r_1^n - c_2 r_1 r_2^n . \end{aligned}$$

Here $r_1 + r_2 = a > 0$, hence $|r_1| > |r_2|$ and $\lim_n \frac{x_n}{y_n} = -r_2 > 0$. From $y_{2n} f(x) + x_{2n} x \downarrow$ and $y_{2n+1} f(x) + x_{2n+1} x \uparrow$ we obtain in the same way that $f(x) - r_2 x \downarrow$ and $f(x) - r_2 x \uparrow$. Hence $f(x) - r_2 x = c$ (constant).

It follows that, in case $r_1 \neq 1$ one must have $c = 0$ and $f = S(r_2)$. In case $r_1 = 1$, all constants $c \in \mathbb{R}$ will do, so the solutions are $S(r_2) + c$, $c \in \mathbb{R}$. $\square$

The following result shows that all solutions must be bi-lipschitzian

Theorem 1.4 (Calibration Theorem)**.** *Let us assume that f is a solution and $|r_1| \leq |r_2|$. Then, for any real x and y one has:*

$$|r_1|\,|x - y| \leq |f(x) - f(y)| \leq |r_2|\,|x - y|.$$

Proof. The result holds in case $r_1 r_2 < 0$, as we have seen in Theorem 1.3. So, let us assume that $r_1 r_2 > 0$.

A. *Case* $0 < r_1 \leq r_2$

We shall write the fundamental equation in the form

$$f \circ f - af + bx = 0 \Leftrightarrow f \circ f = af - bx \uparrow \tag{1.11}$$

with $a > 0$, $b > 0$. In this case $af = f \circ f + bx \uparrow$, hence $f \uparrow$.

We shall find two real sequences $(x_n)_n$, $(y_n)_n$ such that, for any $n \geq 1$ one has

$$x_n f(x) - y_n x \uparrow \tag{1.12}$$

As we have seen, we can start with $x_1 = a > 0$ and $y_1 = b > 0$. From (1.11) we get $af = f \circ f + bx \uparrow$ hence $f \uparrow$ and consequently $af \circ f - bf = a(af - bx) - bf = \left(a^2 - b\right) f - abx \uparrow$. We take $x_2 = a^2 - b > 0$ (because $a^2 - 4b > 0$) and $y_2 = ab$. Assuming (1.12) we obtain

$$\begin{aligned} x_{n+1}f - y_{n+1}x &= x_n f \circ f - y_n f = x_n(af - bx) - y_n f \\ &= (ax_n - y_n)f - bx_n x \uparrow \end{aligned}$$

which gives $x_{n+1} = ax_n - y_n$ and $y_{n+1} = bx_n = r_1 r_2 x_n$ hence $x_{n+2} - ax_{n+1} + bx_n = 0$.

In case $r_1 \neq r_2$, we have $x_n = c_1 r_1^n + c_2 r_2^n$ with

$$\begin{aligned} &c_1 r_1 + c_2 r_2 = a = r_1 + r_2 \\ &c_1 r_1^2 + c_2 r_2^2 = a^2 - b = r_1^2 + r_1 r_2 + r_2^2 \end{aligned}$$

hence $c_1 = \frac{r_1}{r_1 - r_2}$, $c_2 = \frac{r_2}{r_2 - r_1}$ which gives:

$$x_n = \frac{r_2^{n+1} - r_1^{n+1}}{r_2 - r_1} > 0 \quad \text{and} \quad y_n = \frac{r_1 r_2}{r_2 - r_1}\left(r_2^n - r_1^n\right) > 0.$$

We get $\lim_n \frac{x_n}{y_n} = \frac{1}{r_1} > 0$.

In case $r_1 = r_2 = r$, we have $x_n = c_1 r^n + c_2 n r^n$ with

$$\begin{aligned} &(c_1 + c_2)r = a = 2r \\ &c_1 r^2 + 2c_2 r^2 = a^2 - b = 3r^2 \end{aligned}$$

hence $c_1 = c_2 = 1$ which gives $x_n = r^n(1 + n) > 0$ and $y_n = bx_{n-1} = r^2 \cdot r^{n-1} \cdot n = nr^{n+1}$. We get $\lim_n \frac{x_n}{y_n} = \frac{1}{r} > 0$. So, in all cases $x_n > 0$, $y_n > 0$ and $\lim_n \frac{x_n}{y_n} = \frac{1}{r_1}$. We have $x_n f - y_n x \uparrow \Rightarrow \frac{x_n}{y_n} f - x \uparrow$ and, by passing to $n-$ limit

$$\frac{1}{r_1} f - x \uparrow \Rightarrow f - r_1 x \uparrow . \tag{1.13}$$

The fundamental equation for f^{-1} is $f^{-2} - \frac{a}{b} f^{-1} + \frac{1}{b} x = 0$ with characteristic equation $x^2 - \frac{a}{b} x + \frac{1}{b} = 0$ having solutions $\frac{1}{r_2} \leq \frac{1}{r_1}$. According to the previous result

$$f^{-1}(x) - \frac{1}{r_2} x \uparrow \Rightarrow r_2 y - f(y) \uparrow \Rightarrow f(y) - r_2 y \downarrow \tag{1.14}$$

(writing $x = f(y)$).

From (1.13) and (1.14), we obtain for $x > y$

$$r_1(x-y) \le f(x) - f(y) \le r_2(x-y). \tag{I}$$

B. *Case* $r_2 \le r_1 < 0$

The fundamental equation is

$$f \circ f + af + bx = 0 \Leftrightarrow f \circ f = -af - bx \uparrow \tag{1.15}$$

with $a = -r_1 - r_2 > 0$ and $b = r_1 r_2 > 0$. Hence $-af = f \circ f + bx \Rightarrow f \downarrow$. We shall find two sequences $(x_n)_n$, $(y_n)_n$ such that, for all $n \ge 1$

$$(-1)^n x_n f(x) + (-1)^n y_n x \downarrow \tag{1.16}$$

We can start with $x_1 = -a < 0$, $y_1 = -b < 0$ (see (1.15). Also from (1.15) we get $af + bx \downarrow \Rightarrow af \circ f + bf = (-a^2 + b)f - abx \uparrow \Rightarrow (a^2 - b)f(x) + abx \downarrow$. Consequently, we can take $x_2 = a^2 - b > 0$ and $y_2 = ab > 0$. Accepting (1.16) we obtain

$$\begin{aligned}
&(-1)^{n+1} x_{n+1} f(x) + (-1)^{n+1} y_{n+1} x = (-1)^n x_n f \circ f + (-1)^n y_n f \\
&= (-1)^n x_n(-af - bx) + (-1)^n y_n f = (-1)^{n+1} x_n (af + bx) + (-1)^n y_n f \uparrow
\end{aligned}$$

hence $(-1)^{n+1} x_n(-af - bx) + (-1)^{n+1} y_n f \downarrow \Leftrightarrow (-1)^{n+1}(-ax_n + y_n) f + (-1)^{n+1}(-bx_n)x. \downarrow$ We conclude that $x_{n+1} = -ax_n + y_n$ and $y_{n+1} = -bx_n$. We get $x_{n+2} + ax_{n+1} + bx_n = 0$.

As previously, in case $r_1 \ne r_2$ we get $x_n = \frac{r_2^{n+1} - r_1^{n+1}}{r_2 - r_1}$ hence $x_{2n} > 0$, $x_{2n+1} < 0 \Rightarrow y_{2n} > 0$, $y_{2n+1} < 0$. From $x_n = c_1 r_1^n + c_2 r_2^n$, $y_n = -c_1 r_1^n r_2 - c_2 r_2^n r_1$ we obtain $\lim_n \frac{x_n}{y_n} = -\frac{1}{r_1} > 0$.

In case $r_1 = r_2 = r$, we get $x_n = c_1 r^n + n c_2 r^n$, $y_n = -r^2(c_1 r^{n-1} + c_2(n-1)r^{n-1})$ and $\lim_n \frac{x_n}{y_n} = -\frac{1}{r} > 0$. So, $\lim_n \frac{x_n}{y_n} = -\frac{1}{r_1} > 0$ in all cases.

We have successively, with same procedures

$$x_{2n} f + y_{2n} x \downarrow \Rightarrow \frac{x_{2n}}{y_{2n}} f + x \downarrow \Rightarrow -\frac{1}{r_1} f + x \downarrow \Rightarrow f - r_1 x. \downarrow$$

In the same way: $f - r_2 x \uparrow$. Using the last two results for $x > y$:

$$\begin{aligned}
&r_2(x-y) \le f(x) - f(y) \le r_1(x-y) \\
&\Rightarrow -r_1(x-y) \le f(y) - f(x) \le -r_2(x-y)
\end{aligned} \tag{II}$$

From (I) and (II) we get the final result. □

Theorem 1.5 (Case $\Delta = 0$)**.** *Let r be the double solution of the characteristic equation. In case $r \neq 1$, the unique solution is $S(r)$. In case $r = 1$, all the solutions are $S(1) + C$, $C \in \mathbb{R}$.*

Proof. According to the Calibration Theorem 1.4, for any real x one has

$$|r||x - 0| = |f(x) - f(0)|. \tag{1.17}$$

a) In case $r > 0$, it follows that $f \uparrow$ and (1.17) becomes $f(x) - f(0) = rx$, for $x \geq 0$, $f(0) - f(x) = -rx$, for $x < 0$ which means $f(x) = rx + C$, where $C = f(0)$, for all x. If $r = r_1 = r_2 \neq 1$, we already know that the unique possibility is $C = 0$. If $r = r_1 = r_2 = 1$, all functions given via $f(x) = x + C$, $C \in \mathbb{R}$ are solutions.

b) In case $r < 0$, it follows that $f \downarrow$ and (1.17) becomes $f(0) - f(x) = -rx$, for $x \geq 0$, $f(x) - f(0) = rx$, for $x < 0$ which means $f(x) = rx + C$, where $C = f(0)$. Because $r = r_1 = r_2 \neq 1$, only $C = 0$ will furnish a solution. $\square$

As a conclusion of this first part, we can see that the cases which remain to be studied are those when *the roots r_1, r_2 are different real numbers having the same sign.*

In this case $b = r_1 r_2 > 0$ and $a = -(r_1 + r_2)$. One can write the fundamental equation in the form $-af(x) = f \circ f(x) + bf(x) \uparrow$. This shows that in case $a > 0$ (i.e. r_1, r_2 are negative) a solution $f \downarrow$ and in case $a < 0$ (i.e. r_1, r_2 are positive) a solution $f \downarrow$.

2. Solving the functional equation: the remaining cases

Theorem 2.1 (Case $r_1 = 1 < r_2 \Leftrightarrow 0 < r_2 < r_1 = 1$)**.** *The solutions are of the following four types:*

a) $f = S(1) = 1_{\mathbb{R}}$*;*

b) $f = S(r_2) + C$, $C \in \mathbb{R}$.

c) There exists $x_1 \in \mathbb{R}$ such that:

$$f(x) = \begin{cases} x, & \text{if } x < x_1 \\ r_2 x + c, & \text{if } x \geq x_1 \end{cases}$$

with $r_2 x_1 + c = x_1$.

d) There exists $x_1 \in \mathbb{R}$ *such that*

$$f(x) = \begin{cases} r_2 x + c, & \text{if } x < x_1 \\ x, & \text{if } x \geq x_1 \end{cases}$$

with $r_2 x_1 + c = x_1$.

e) There exist real $x_1 < y_1$ *such that*

$$f(x) = \begin{cases} r_2 x + c_1, & \text{if } x < x_1 \\ x, & \text{if } x_1 \leq x \leq y_1 \\ r_2 x + c_2, & \text{if } x > y_1 \end{cases}$$

with $r_2 x_1 + c_1 = x_1$ *and* $r_2 y_1 + c_2 = y_1$.

Proof. An examination of a), b), c), d), e) shows that the exhibited functions are, indeed, solutions.

Now, let us consider a solution $f : \mathbb{R} \to \mathbb{R}$, which satisfies the fundamental equation $f \circ f - (r_2 + 1)f(x) + r_2 x = 0$ which can be written as follows

$$(r_2 + 1)f(x) = f \circ f(x) + r_2 x \Rightarrow f(f(x)) - r_2 f(x) = f(x) - r_2 x. \tag{2.1}$$

From the first part of (2.1) it follows that $f \uparrow$.

We introduce the function $h : \mathbb{R} \to \mathbb{R}$, given via $h(x) = f(x) - r_2 x$. Using the Calibration Theorem 1.4 we get, for $x > y : x - y \leq f(x) - f(y) \leq r_2(x - y)$ which shows that $h \downarrow$.

Our next aim is to show that the set $fix(f)$ is not empty. Indeed, if $fix(f) = \emptyset$, one must have either $f(x) > x$, or $f(x) < x$, for all real x. Assume, e. g. that $f(x) > x$ for all x (the proof in the other case is similar). Fixing an $x_0 \in \mathbb{R}$, we obtain the strictly increasing sequence $(x_n)_{n \in \mathbb{Z}}$, given via $x_{n+1} = f(x_n)$ and $x_{-n-1} = f^{-1}(x_{-n})$, for al natural n.

One has $L = \lim_{n \to \infty} x_n = \infty$ and $l = \lim_{n \to -\infty} x_n = -\infty$.

Indeed, if $L < \infty$ or $l > -\infty$, one uses the continuity of f or of f^{-1} to get $L = f(L)$ or $l = f^{-1}(l)$, which is impossible. It follows that $\mathbb{R}$ can be written as the reunion of all intervals $[f^{n-1}(x_0), f^n(x_0)]$, $n \in \mathbb{Z}$.

Now we use the second part of (2.1) in the form $h(f(x)) = h(x)$ to obtain for any real x and for any $n \in \mathbb{Z}$

$$h\left(f^n(x)\right) = h(x). \tag{2.2}$$

It follows that, for all $n \in \mathbb{Z}$, one has $h\left(f^{n-1}(x_0)\right) = h\left(f^n(x_0)\right) = h(x_0)$. Using $h \downarrow$, it follows that h is constant on $\left[f^{n-1}(x_0), f^n(x_0)\right]$. If c_n is this constant, it follows that all c_n must be equal, because h is continuous. Hence, there exists a real constant c such that $h(x) = c$ for all real x, which means

$$f(x) = r_2 x + c \tag{2.3}$$

for all real x. We arrived at a contradiction because, from (2.3) one can solve the equation $f(x) = x \Leftrightarrow x = \frac{c}{1-r_2}$ and this contradicts the fact that $fix(f) = \emptyset$.

Because $fix(f)$ is closed, let us write $\mathbb{R} \setminus fix(f) = \bigcup_{n \in P}(x_n, y_n)$ where P is at most countable and the intervals $I_n = (x_n, y_n)$ are non empty and mutually disjoint.

We remark that all I_n must be unbounded. Indeed, assume $I_n = (x_n, y_n)$ is bounded, i. e. x_n and y_n are numbers. It follows that $f(x_n) = x_n$, $f(y_n) = y_n$ and e. g. $f(x) > x$ for all $x \in I_n$. This shows that $f(I_n) \subset I_n$ and, for any $x \in I_n$, the sequence $(f^m(x))_{m \in \mathbb{Z}}$ is strictly increasing with

$$\lim_{m \to \infty} f^m(x) = y_n \quad \text{and} \quad \lim_{m \to -\infty} f^m(x) = x_n. \tag{2.4}$$

For such $x \in I_n$ we have (see (2.2) and (2.4)) $h(x) = h(f^m(x)) = h(x_n) = h(y_n)$ which shows that h is constant on I_n. This implies the existence of $c_n \in \mathbb{R}$ such that, for all $x \in (x_n, y_n)$, $f(x) = r_2 x + c_n$. But $f(x_n) = x_n$, $f(y_n) = y_n$ and the continuity of f implies $r_2 x_n + c_n = x_n$ and $r_2 y_n + c_n = y_n \Rightarrow r_2(y_n - x_n) = y_n - x_n$ which is impossible because $r_2 > 1$.

The conclusion is that all intervals I_n must be unbounded. So, there are at most two intervals I_n.

First case: 0 intervals I_n. In this case $fix(f) = \mathbb{R}$ and we have case a) in the enunciation.

Second case: 1 interval I_n. This I_n can be of the form $(-\infty, x_1)$ and we have case d) or of the form (x_1, ∞) and we have case c).

Third case: 2 intervals I_n. These I_n can be of the form $(-\infty, x_1)$ and (x_1, ∞) and we have case b) or of the form $(-\infty, x_1)$ and (y_1, ∞), with $x_1 < y_1$ and we have case e). □

Remark. It is seen that in the equivalent case $0 < r_2 < r_1 = 1$, the solutions have the some form.

Lemma 2.2. *Let us assume* $1 < r_1 < r_2$ $(\Leftrightarrow 0 < r_1 < r_2 < 1)$. *For any solution* f *we have the following properties:*

a) $f(0) = 0$

b) For any $x_0 \in \mathbb{R}$ *one has*

$$\begin{cases} r_1 x_0 \le f(x_0) \le r_2 x_0, & \text{if } x_0 \ge 0 \\ r_2 x_0 \le f(x_0) \le r_1 x_0, & \text{if } x_0 < 0. \end{cases}$$

Proof. a) Assume $f(0) > 0$. Applying the Calibration Theorem 1.4 to $f \uparrow$ we obtain $r_1 f(0) \le f(f(0)) - f(0) \le r_2 f(0)$.

Because $f(f(0)) = (r_1 + r_2) f(0)$ we get $r_1 f(0) \le (r_1 + r_2 - 1) f(0) \le r_2 f(0)$, hence $r_1 \le r_1 + r_2 - 1 \le r_2 \Leftrightarrow r_2 \ge 1$ and $r_1 \le 1$ which is a contradiction.

In case $f(0) < 0$ we get $r_1 (-f(f(0)) \le f(0) - f(f(0)) \le r_2(-f(0))$ which gives $r_1 \le r_1 + r_2 - 1 \le r_2$ a.s.o.

b) Again the Calibration Theorem gives $r_1(x_0 - 0) \le f(x_0) - f(0) \le r_2(x_0 - 0)$, for $x_0 \ge 0$, $r_1(0 - x_0) \le f(0) - f(x_0) \le r_2(0 - x_0)$, for $x_0 < 0$ a.s.o. $\square$

Lemma 2.3. *Let us assume that* $r_2 < r_1 < 0$. *For any solution* f *we have the following properties:*

a) $f(0) = 0$,

b) For any $x_0 \in \mathbb{R}$ *one has:*

$$\begin{cases} r_2 x_0 \le f(x_0) \le r_1 x_0, & \text{if } x_0 \ge 0 \\ r_1 x_0 \le f(x_0) \le r_2 x_0, & \text{if } x_0 < 0. \end{cases}$$

Proof. a) Assume $f(0) > 0$. Again Theorem 1.4 applied to $f \downarrow$ gives $-r_1(f(0) - 0) \le f(0) - f(f(0)) \le -r_2(f(0) - 0)$ hence $-r_1 \le 1 - r_1 - r_2 \le -r_2 \Rightarrow 1 - r_1 \le 0$ false.

Assume $f(0) < 0$. We get $-r_1(0 - f(0)) \le f(f(0)) - f(0) \le -r_2(0 - f(0)) \Rightarrow r_1 f(0) \le (r_1 + r_2 - 1) f(0) \le r_2 f(0) \Rightarrow r_1 \ge r_1 + r_2 - 1 \ge r_2 \Rightarrow r_1 - 1 \ge 0$, false. $\square$

Lemma 2.4. *Let us assume that* $r_2 < r_1 < -1$ $(\Leftrightarrow -1 < r_1 < r_2 < 0)$. *Let* $0 \ne x_0 \in \mathbb{R}$ *and* $x_1 \in [r_2 x_0, r_1 x_0]$ *(in case* $x_0 > 0$*) or* $x_1 \in [r_1 x_0, r_2 x_0]$ *(in case* $x_0 < 0$*). Using the coefficients of the fundamental equation* (1.1) *we define the sequence* $(x_n)_{n \ge 0}$ *and* $(x_{-n})_{n \ge 0}$ *as follows:*

a) $x_{n+2} = -a x_{n+1} - b x_n$, *with starting terms* x_0 *and* x_1.

Such a sequence is the sequence given via $x_{n+1} = f(x_n)$*, with starting term* x_0 *(see Lemma* 2.3 *for* $x_1 = f(x_0)$*).*

b) The sequence $(x_{-n})_n$ *is defined in two steps:*

Firstly we define the sequence $(y_n)_{n\geq 0}$ *given via (see* (1.4)*)* $y_{n+2} = -\frac{a}{b}y_{n+1} - \frac{1}{b}y_n$ *with starting terms* $y_0 = x_1$ *and* $y_1 = x_0$*.*

Next we write $x_{-m}=y_{m+1}$*, for all natural* m*. Hence* $x_{-m-2}=-\frac{a}{b}x_{-m-1}-\frac{1}{b}x_{-m}$ *with starting terms* $x_0 = y_1$ *and* $x_{-1} = y_2$*. Such a sequence is the sequence given via* $x_{-n-1} = f^{-1}(x_n)$*, with starting term* x_0 *(see Lemma* 2.3 *for* $x_1 = f(x_0)$*, hence* $x_0 = f^{-1}(x_1)$*).*

In case $x_0 > 0$*, we have:* $x_{2n} \uparrow \infty$ *(strictly),* $x_{2n+1} \downarrow -\infty$ *(strictly),* $x_{-2n} \downarrow 0$ *(strictly) and* $x_{-2n+1} \uparrow 0$ *(strictly). This implies*

$$\bigcup_{n\geq 0} ([x_{2n}, x_{2n+2}] \cup [x_{-2n}, x_{-2n+2}]) = (0,\infty)$$

$$\bigcup_{n>0} ([x_{2n+1}, x_{2n-1}] \cup [x_{-2n+1}, x_{-2n-1}]) = (-\infty, 0).$$

The case $x_n < 0$ *is symmetric (e.g.* $x_{2n} \downarrow -\infty$ *strictly* $\ldots$*).*

Proof. We shall be concerned with the case $x_0 > 0$, the case $x_0 < 0$ being similar.

a) We have

$$x_n = c_1 r_1^n + c_2 r_2^n \tag{2.5}$$

with $c_1 + c_2 = x_0$, $c_1 r_1 + c_2 r_2 = x_1$ hence

$$c_1 = \frac{x_1 - r_2 x_0}{r_1 - r_2} \geq 0, \qquad c_2 = \frac{r_1 x_0 - x_1}{r_1 - r_2} \geq 0 \tag{2.6}$$

and c_1, c_2 cannot be null at the same time.

The form of x_n proves everything.

b) To prove the assertions oncerning $(x_{-n})_n$, we shall be concerned with $(y_n)_n$. We have

$$y_n = b_1 \left(\frac{1}{r_1}\right)^n + b_2 \left(\frac{1}{r_2}\right)^n \tag{2.7}$$

with $b_1 + b_2 = x_1$, $b_1 \frac{1}{r_1} + b_2 \frac{1}{r_2} = x_0$ hence

$$b_1 = \frac{r_1(x_1 - r_2 x_0)}{r_1 - r_2} \leq 0, \qquad b_2 = \frac{r_2(r_1 x_0 - x_1)}{r_1 - r_2} \leq 0 \tag{2.8}$$

and c_1, c_2 cannot be both null.

It follows that $y_{2k} \uparrow 0$ and $y_{2k+1} \downarrow 0$ strictly. $\square$

Lemma 2.5. *Let us assume that* $1 < r_1 < r_2$ ($\Leftrightarrow 0 < r_1 < r_2 < 1$). *Let* $0 \neq x_0 \in \mathbb{R}$ *and* $x_1 \in [r_1x_0, r_2x_0]$ (*in case* $x_0 > 0$) *or* $x_1 \in [r_2x_0, r_1x_0]$ (*in case* $x_0 < 0$). *We define the sequences* $(x_n)_{n\geq 0}$ *and* $(x_{-n})_{n\geq 0}$ *exactly like in Lemma* 2.4.

In particular, we can take $x_{n+1} = f(x_n)$, *with starting term* x_0 *and* $x_{-n-1} = f^{-1}(x_{-n})$, *with starting term* x_0 (*see Lemma 2.2*).

In case $x_0 > 0$, *we have:* $x_n \uparrow \infty$ (*strictly*), $x_{-n} \downarrow 0$ (*strictly*). *In case* $x_0 < 0$, *we have* $x_n \downarrow -\infty$ (*strictly*) *and* $x_{-n} \uparrow 0$ (*strictly*). *This implies*

$$\bigcup_{n\in\mathbb{Z}} [x_n, x_{n+1}] = (0, \infty) \ \ if \ \ x_0 > 0$$
$$\bigcup_{n\in\mathbb{Z}} ([x_n, x_{n+1}] = (-\infty, 0) \ \ if \ \ x_0 < 0.$$

Proof. Formulae (2.5), (2.6), (2.7) and (2.8) hold also here. In case $x_0 > 0$, it is seen that $c_1 \geq 0$, $c_2 \geq 0$ and $b_1 \geq 0$, $b_2 \geq 0$, whereas in case $x_0 < 0$, it is seen that $b_1 \leq 0$, $b_2 \leq 0$ and everything follows. $\square$

Lemma 2.6. *Let us assume that* $0 < r_1 < 1 < r_2$. *Let* $x_0 \in \mathbb{R}$ *and* $x_1 > r_1x_0$, $x_1 > r_2x_0$. *We define the sequence* $(x_n)_{n\geq 0}$ *and* $(x_{-n})_{n\geq 0}$ *exactly like in Lemma* 2.4.

In particular, we can take $x_{n+1} = f(x_n)$, *with starting term* x_0 *and* $x_{-n-1} = f^{-1}(x_{-n})$, *with starting term* x_0. *Then* $x_n \uparrow \infty$ (*strictly*), $x_{-n} \downarrow -\infty$ (*strictly*) *and* $\bigcup_{n\in\mathbb{Z}} [x_n, x_{n+1}] = \mathbb{R}$.

Proof. Formulae (2.5), (2.6), (2.7) and (2.8) are valid also here. One can see that $c_2 \geq 0$, $c_1 \leq 0$, hence $x_n = c_1r_1^n + c_2r_2^n \underset{n}{\to} \infty$. Besides: $x_{n+1} - x_n = c_1r_1^n(r_1 - 1) + c_2r_2^n(r_2 - 1) > 0$.

On the other hand $b_1 < 0$, $b_2 > 0$, hence $y_n = b_1 \left(\frac{1}{r_1}\right)^n + b_2 \left(\frac{1}{r_2}\right)^n \underset{n}{\to} \infty$. We have also

$$y_{n+1} - y_n = b_1 \left(\frac{1}{r_1}\right)^n \left(\frac{1}{r_1} - 1\right) + b_2 \left(\frac{1}{r_2}\right)^n \left(\frac{1}{r_2} - 1\right)^n < 0.$$

$\square$

Theorem 2.7 (Case $r_2 < -1 < r_1 < 0$)**.** *The solutions are* $S(r_1)$ *and* $S(r_2)$.

Proof. Let f be a solution. We have $f \downarrow$ and $f(0) = 0$ (Lemma 2.3). We define the functions:

$$g : \mathbb{R} \to \mathbb{R}, \quad g(x) = f(x) - r_1 x$$
$$h : \mathbb{R} \to \mathbb{R}, \quad h(x) = f(x) - r_2 x$$

and one sees that $g \downarrow$ and $h \uparrow$. (The Calibration Theorem 1.4). The identities $g(f(x)) = r_2 g(x)$ and $h(f(x)) = r_1 h(x)$ furnish the identies (valid for any real x and any natural n, using mathematical induction):

$$g\left(f^{2n}(x)\right) = r_2^{2n} g(x) \quad \text{and} \quad h\left(f^{2n}(x)\right) = r_1^{2n} h(x) \tag{2.9}$$

(e.g. $g(f(y)) = r_2 g(y) \Rightarrow$ for $y = f(x) : g\left(f^2(x)\right) = r_2 g\left(f(x)\right) = r_2^2 g(x)$; $g\left(f^{2n+2}(x)\right) = g\left(f^2\left(f^{2n}(x)\right)\right) = r_2^2 g\left(f^{2n}(x)\right) = r_2^{2+2n} g(x)$ a.s.o.)

Let $x_0 > 0$. One has $f \circ f(x_0) > f \circ f(0) = 0$. Accepting that $x_{2n} = f^{2n}(x_0) > 0$ we obtain $f^{2n+2}(x_0) = f \circ f\left(f^{2n}(x_0)\right) > 0$ and (x_{2n}) has only strictly positive terms.

Using Lemma 2.3 for $f(x_0) < 0$, one has $r_1 f(x_0) \leq f\left(f\left(x_0\right)\right)$ and $f(x_0) \leq r_1 x_0 \Rightarrow r_1 f(x_0) \geq r_1^2 x_0 \Rightarrow r_1^2 x_0 \leq f^2(x_0) \Rightarrow r_1^2 x_0 \leq x_2$.

Mathematical induction gives $x_{2n} \geq r_1^{2n} x_0$ for all natural n (e.g. $x_{2n} \geq r_1^{2n} x_0 \Rightarrow x_{2n+2} \geq f \circ f\left(r_1^{2n} x_0\right) \geq r_1^2 \cdot r_1^{2n} x_0$ a. s. o.). One has $f^2(x_n) - (r_1 + r_2) f(x_n) + r_1 r_2 x_n = x_{n+2} - (r_1 + r_2) x_{n+1} + r_1 r_2 x_n = 0$ which shows that there exist constants c_1, c_2 such that $c_1 = \frac{r_2 x_0 - x_1}{r_2 - r_1}$, $c_2 = \frac{x_1 - r_1 x_0}{r_2 - r_1}$.

$$x_{2n} = c_1 r_1^{2n} + c_2 r_2^{2n}. \tag{2.10}$$

From (2.10) it follows that there exists $\lim_n x_{2n} \in \{0, \infty\}$.

There are two possibilities: either $x_0 < x_2$ (and, via induction, $x_{2n} \uparrow$ strictly) and $x_{2n} \to \infty$ or $x_0 > x_2$ (and , via induction, $x_{2n} \downarrow 0$ strictly) and $x_{2n} \to 0$. The case $x_0 = x_2$ is not possible, because the sequence $(x_{2n})_n$ cannot be constant, in view of (2.10) and of the form of the coefficients c_1, c_2 which cannot be both null.

We have the equivalence

$$x_{2n} = f^{2n}(x_0) \underset{n}{\to} \infty \Leftrightarrow f^{-2n}(x_0) \underset{n}{\to} 0 \tag{2.11}$$

(using the facts that $x_{2n} \uparrow$ strictly $\Leftrightarrow$ $x_0 < x_2 \Leftrightarrow 0 < x_0 < f^2(x_0) \Leftrightarrow 0 < f^{-2}(x_0) < x_0 \Leftrightarrow f^{-2n}(x_0) \downarrow$ strictly and, supplementary, (2.10) for $\frac{1}{r_1}$ and $\frac{1}{r_2}$).

In case $x_0 < 0$ one can prove that $f^{2n}(x_0) = x_{2n} < 0$, $x_{2n} \leq r_1^{2n} x_0$ for all n and $\lim_n x_{2n} \in \{-\infty, 0\}$.

Moreover, we have the equivalence

$$f^{2n}(x_0) \underset{n}{\to} -\infty \Leftrightarrow f^{-2n}(x_0) \underset{n}{\to} 0. \tag{2.12}$$

For any $0 \neq x_0 \in \mathbb{R}$ we have (see (2.9)):

$$g(x_0) = \frac{g\left(f^{2n}(x_0)\right)}{r_2^{2n}}. \tag{2.13}$$

$$h(x_0) = h\left(f^{2n}\left(f^{-2n}(x_0)\right)\right) = r_1^{2n} h\left(f^{-2n}(x_0)\right). \tag{2.14}$$

We use (2.11), (2.12), (2.13) and (2.14).

In case $f^{2n}(x_0) \to 0$, because $\frac{1}{r_2^{2n}} \underset{n}{\to} 0$, we obtain $g(x_0) = 0$. In case $f^{2n}(x_0) \underset{n}{\to} \infty$ (or $f^{2n}(x_0) \underset{n}{\to} -\infty$) it follows that $h(x_0) = 0$, because $r_1^{2n} \to 0$ and $f^{-2n}(x_0) \to 0$.

The conclusion is the following: for any $0 \neq x_0 \in \mathbb{R}$ one has either $g(x_0) = 0 \Leftrightarrow f(x_0) = r_1 x_0$ or $h(x_0) = 0 \Leftrightarrow f(x_0) = r_2 x_0$ (this is true also for $x_0 = 0$).

We shall show that one must have either $f(x) = r_1 x$ for all x or $f(x) = r_2 x$ for all x.

Indeed: there are two possibilities: either $\lim_n f^{2n}(x_0) = 0$ for all $0 \neq x_0 \in \mathbb{R}$ (and in this case $g(x_0) = 0 \Leftrightarrow f(x_0) = r_1 x_0$ for all $x_0 \in \mathbb{R}$) or there exists $0 \neq x_0 \in \mathbb{R}$ such that $\lim_n f^{2n}(x_0) = \infty$ (or $\lim_n f^{2n}(x_0) = -\infty$). We shall show that in this case one most have $h(x) = 0$, i.e. $f(x) = r_2 x$ for all x.

Assume, e.g. the existence of $x_0 > 0$ such that $\lim_n f^{2n}(x_0) = \infty$.

It follows from (2.14) that $h(x_0) = 0$. For all $x_0' > x_0$ one has $f^{2n}(x_0') \geq f^{2n}(x_0)$, hence $\lim_n f^{2n}(x_0') = \infty$ and $h(x_0') = 0$. Hence,

$$f(x) = r_2 x, \quad x \in [x_0, \infty). \tag{2.15}$$

Put $x_1 = f^{-1}(x_0) < 0$. We have $h(x_0) = r_1^{2n} h\left(f^{-2n}(x_0)\right)$, which means $h\left(f(x_1)\right) = r_1^{2n} h\left(f^{-2n}(x_0)\right)$. Because $h\left(f(x_1)\right) = r_1 h(x_1)$, we get $r_1 h(x_1) = r_1^{2n} h\left(f^{-2n}(x_0)\right) \underset{n}{\to} 0$ and this implies $h(x_1) = 0$. Now, let us take $x_1' \leq x_1$. We have $x_1' = f^{-1}\left(f\left(x_1'\right)\right) = f^{-1}\left(x_0'\right)$, with $x_0' = f\left(x_0'\right) \geq f(x_1) = x_0$.

Using the fact that $h\left(x_0'\right) = 0$ (see (2.15)) we obtain in a similar manner that $h\left(x_1'\right) = 0$. The conclusion is that

$$f(x) = r_2 x, \qquad x \in (-\infty, x_1] \tag{2.15'}$$

What happens on (x_1, x_0)? Let us define $l = \sup\left\{x \mid x < 0,\ f(x) = r_2 x\right\}$.

In case $l < 0$, we have $f(l) = r_2 l$ (due to the continuity of f and (2.15$'$). Let $(x_n)_n$ be a sequence such that $l < x_n < 0$ and $x_n \to l$. One must have $f(x_n) = r_1 x_n$ for all n, hence passing to $n-$limit: $f(l) = r_1 l$, which is a contradiction, consequently $l = 0$. In the same way, using (2.15):

$$L = \inf\left\{x \mid x > 0,\ f(x) = r_2 x\right\} = 0.$$

Consequently, it follows that $f(x) = r_2 x$ for all $x \in (x_1, x_0)$, hence for all $x \in \mathbb{R}$. $\square$

Theorem 2.8 (Case $r_2 < r_1 = -1 \Leftrightarrow -1 = r_1 < r_2 < 0$)**.** *The solutions are* $S(-1)$ *and* $S(r_2)$.

Proof. We have $f \downarrow$ and $f(0) = 0$, according to Lemma 2.3, for any solution.

We begin with the remark that the unique fixed point of f is $x_0 = 0$. Indeed, if $x_0 \neq 0$ would be a fixed point $x_0(1 + a + b) = 0 \Leftrightarrow 1 + a + b = 0$ which contradicts the fact that $b > 0$ and (see (1.2) $1 - a + b = 0$. The fundamental equation can be rewritten

$$f \circ f + (1 - r_2) f - r_2 x = \left(f \circ f(x) + f(x)\right) - r_2 \left(f(x) + x\right) = 0. \tag{2.16}$$

which means

$$h(f(x)) - r_2 h(x) = 0 \tag{2.17}$$

where $h : \mathbb{R} \to \mathbb{R}$, $h(x) = f(x) + x$. We have also

$$g(f(x)) + g(x) = 0 \tag{2.18}$$

where $g : \mathbb{R} \to \mathbb{R}$, $g(x) = f(x) - r_2 x$.

The Calibration Theorem gives for $x > y$, $x - y \leq f(y) - f(x) \leq -r_2(x - y)$ and we get $g \uparrow$ and $h \downarrow$.

There are two possibilities: either $h(x) = 0$ for all $x \in \mathbb{R}$ (which means $f(x) = -x$ for all $x \in \mathbb{R}$), or there exists $x \in \mathbb{R}$ such that $h(x_0) \neq 0$.

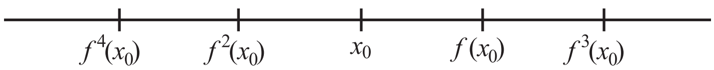
Let us examine the second possibility. In any case $x_0 \neq 0$ and x_0 cannot be a fixed point for f. Let us assume e.g. that $h(x_0) > 0$, i.e. $f(x_0) > -x_0$ and (supplementary) $f(x_0) > x_0$ (the other cases are treated similarly). From (2.17), with $x = f(y)$, we get $h\left(f^2(y)\right) = r_2 h\left(f(y)\right) = r_2^2 h(y)$ and hence (induction), for all natural n

$$h\left(f^n(x_0)\right) = r_2^n h\left(x_0\right). \tag{2.19}$$

Hence, the sequence $\left(h\left(f^{2n}(x_0)\right)\right)_n$ is strictly increasing.

It follows that the sequence $\left(f^{2n}(x_0)\right)_n$ is strictly decreasing and one must have $\lim_n f^{2n}(x_0) = -\infty$. Indeed, în case $l = \lim_n f^{2n}(x_0) > -\infty$, one has $\lim_n h\left(f^{2n}(x_0)\right) = h(l) \in \mathbb{R}$, contradicting the fact (see (2.19)) that $\lim_n r_2^{2n} h(x_0) = \infty$. In the same way, the sequence $\left(f^{2n+1}(x_0)\right)_n$ is strictly increasing and $\lim_n f^{2n+1}(x_0) = \infty$.

The position on the real axis must be the following:



Indeed, we began with $f(x_0) > x_0$, which implies $f(x_0) > f^2(x_0)$. We cannot have $f^2(x_0) = f(x_0)$ (which implies $f^{2k}(x_0) = f(x_0)$ or $f^2(x_0) = x_0$ (which implies $f^{2k}(x_0) = x_0$). The following possibilities remain:

a) $x_0 < f^2(x_0) = f(x_0)$;

b) $f^2(x_0) < x_0$.

Variant a) gives $f(x_0) > f^3(x_0) > f^2(x_0)$ hence $x_0 < f^2(x_0) < f^3(x_0) < f(x_0)$ and, continuing, one can see that all the values $\left(f^n(x_0)\right)_n$ are between x_0 and $f(x_0)$ and this is not possible. So, only variant b) is acceptable, as we have stated.

From (2.18), we get for $x = f(y)$, $g\left(f^2(y)\right) = -g\left(f(y)\right) = -\left(-g(y)\right)$ and we get for any $n \in \mathbb{N}$

$$g\left(f^n(x_0)\right) = (-1)^n g(x_0). \tag{2.20}$$

From (2.20) we obtain for all n that $g\left(f^{2n}(x_0)\right) = g(x_0)$. Because $g \uparrow$, it follows (we have $\lim_n f^{2n}(x_0) = -\infty$) that g must be constant on all intervals $\left[f^{2k+1}(x_0), f^{2k}(x_0)\right]$, hence g must be constant on $(-\infty, x_0]$.

The first conclusion is that $g(x) = g(x_0)$ for all $x \in (-\infty, x_0]$. This means that $f(x) = r_2 x + g(x_0)$ for all $x \in (-\infty, x_0]$. It follows that, because $r_1 \neq 1$, $r_2 \neq 1$, we have $g(x_0) = 0$ (see the beginning of the paper). Hence:

$$f(x) = r_2 x, \quad \text{for all} \quad x \in (-\infty, x_0]. \tag{2.21}$$

In the same way, for all $n \in \mathbb{N}$, one has $g\left(f^{2n+1}(x_0)\right) = -g(x_0)$ and, consequently $g(x) = -g(x_0) = 0$, for all $\quad x \in [f(x_0), -\infty)$, which means

$$f(x) = r_2 x, \quad \text{for all} \quad x \in [f(x_0), \infty). \tag{2.22}$$

In case $x_0 \geq r_2 x_0$, using (2.21) and (2.22), it follows that $f(x) = r_2 x$ for all $x \in \mathbb{R}$, hence $f = S(r_2)$.

In case $x_0 < r_2 x_0$ (i.e. $x_0 < 0$), we use the idea of the previous proof: we have proved that $h(x_0) \neq 0 \Rightarrow f(x_0) = r_2 x_0$.

We define $l = \sup\{x \mid x < 0, \quad h(x) \neq 0 \Leftrightarrow f(x) = r_2 x\}$. In case $l < 0$, we have $f(l) = r_2 l$ due to the continuity of f. Taking a sequence $l < x_n < 0$ such that $x_n \to l$, one must have $f(x_n) = -x_n$ (because $f(x_n) \neq r_2 x_n$). Hence, $\lim_n f(x_n) = f(l) = -l$ and we got a contradiction and $l = 0$.

In the same way: $L = \inf\{x \mid x > 0, \quad h(x) \neq 0 \Leftrightarrow f(x) = r_2 x\} = 0$. Finally, we get $f(x) = r_2 x$, for all $x \in \mathbb{R}$, hence $f = S(r_2)$. $\square$

Remark. In the equivalent case $-1 = r_1 < r_2 < 0$, the solutions are the same: $S(-1)$, $S(r_2)$.

Theorem 2.9 (Case $1 < r_1 < r_2 \Leftrightarrow 0 < r_1 < r_2 < 1$)**.** *We shall write the fundamental equation in the form $f \circ f - af + bx = 0$. All the solutions $f : \mathbb{R} \to \mathbb{R}$ are of the form*

$$f(x) = \begin{cases} F_1(x), & \text{if } x > 0 \\ 0, & \text{if } x = 0 \\ F_2(x), & \text{if } x < 0, \end{cases}$$

where F_1 and F_2 are constructed as follows:

1. *We construct the sequences $(x_n)_{n \geq 0}$ and $(x_{-n})_{n \geq 0}$ according to Lemma 2.5 starting with an arbitrary $x_0 > 0$ and $x_1 \in [r_1 x_0, r_2 x_0]$. We consider a bijection $f_0 : [x_0, x_1] \to [x_1, x_2]$ having the property that for any $x > y$ in $[x_0, x_1]$ one has*

$$r_1(x - y) \leq f_0(x) - f_0(y) \leq r_2(x - y). \tag{2.23}$$

Then, for any natural n, one can construct the bijections

$$f_n : [x_n, x_{n+1}] \to [x_{n+1}, x_{n+2}] \quad \text{and} \quad f_{-n} : [x_{-n}, x_{-n+1}] \to [x_{-n+1}, x_{-n+2}]$$

defined via

$$f_{n+1}(x) = ax - bf_n^{-1}(x)(x) \quad \text{and} \quad f_{-n+1}^{-1}(x) = \frac{a}{b}x - \frac{1}{b}f_{-n}(x). \tag{2.24}$$

Finally, for any $x \in (0, \infty) = \bigcup_{n \in \mathbb{Z}} [x_n, x_{n+1}]$*, we have, for some natural* n

– *either* $x \in [x_n, x_{n+1}]$ *and* $F_1(x) = f_n(x)$

– *or* $x \in [x_{-n}, x_{-n+1}]$ *and* $F_1(x) = f_{-n}(x)$

(*the values at the common endpoints coincide*).

2. *We construct the sequences* $(x_n)_{n \geq 0}$ *and* $(x_{-n})_{n \geq 0}$ *according to Lemma* 2.5 *starting with an arbitrary* $x_0 < 0$ *and* $x_1 \in [r_2 x_0, r_1 x_0]$*. We consider a bijection* $f_0 : [x_1, x_0] \to [x_2, x_1]$ *having the property* (2.23) *for any* $x > y$ *in* $[x_1, x_0]$.

Then, for any natural n*, one can construct the bijections*

$$f_n : [x_{n+1}, x_n] \to [x_{n+2}, x_{n+1}] \quad \textit{and} \quad f_{-n} : [x_{-n+1}, x_{-n}] \to [x_{-n+2}, x_{-n+1}]$$

defined via (2.24).

Finally, for any $x \in (-\infty, 0) = \bigcup_{n \in \mathbb{Z}} [x_{n+1}, x_n]$*, we have, for some natural* n

– *either* $x \in [x_{n+1}, x_n]$ *and* $F_2(x) = f_n(x)$

– *or* $x \in [x_{-n+1}, x_{-n}]$ *and* $F_2(x) = f_{-n}(x)$

(*the values at the common endpoints coincide*).

Proof. We begin with the remark that functions f_0 with the stipulated property do exists. For instance, in case $x_0 > 0$, one can take $f_0(x) = mx + n$, with $f_0(x_0) = x_1$, $f_0(x_1) = x_2$ and we get $r_1 \leq m = \frac{x_2 - x_1}{x_1 - x_0} \leq r_2$, a.s.o.

A. We construct the functions $F_1 : (0, \infty) \to \mathbb{R}$ and $F_2 : (-\infty, 0) \to \mathbb{R}$ showing how the bijections in the enunciation appear.

The construction of F_1 begins with $x_0 > 0$ and the initial strictly increasing bijections $f_0 : [x_0, x_1] \to [x_1, x_2]$ which has the property (2.23). This is the first (number 0) step of the induction which will prove that the functions f_n and f_{-n} (see the enunciation) are strictly increasing bijections having the properties that, for all $x > y$ in their domains

$$\begin{aligned} r_1(x-y) \leq f_n(x) - f_n(y) \leq r_2(x-y) \quad \text{and} \\ r_1(x-y) \leq f_{-n}(x) - f_{-n}(y) \leq r_2(x-y). \end{aligned} \tag{2.25}$$

Let us accept this hypothesis (number n). We proceed to step $n+1$.

Defining $u : [x_{n+1}, x_{n+2}] \to \mathbb{R}$ via $u(x) = ax - bf_n^{-1}(x)$ we get $u(x_{n+1}) = x_{n+2}$ and $u(x_{n+2}) = x_{n+3}$.

From the hypothesis (2.25) we obtain, taking $f_n^{-1}(x)$ and $f_n^{-1}(y)$ (for $x > y$ in $[x_{n+1}, x_{n+2}]$) instead of x and y

$$\frac{1}{r_2}(x-y) \leq f_n^{-1}(x) - f_n^{-1}(y) \leq \frac{1}{r_1}(x-y). \tag{2.26}$$

Because $u(x) - u(y) = a(x-y) - b\left(f_n^{-1}(x) - f_n^{-1}(y)\right)$ we get from (2.26) $a(x-y) - \frac{b}{r_1}(x-y) \le u(x) - u(y) \le a(x-y) - \frac{b}{r_2}(x-y)$. But $a = r_1 + r_2$ and $b = r_1 r_2$, and the last double inequality becomes:

$$r_1(x-y) \le u(x) - u(y) \le r_2(x-y). \tag{2.27}$$

At this moment we proved that the function $f_{n+1} : [x_{n+1}, x_{n+2}] \to [x_{n+2}, x_{n+3}]$ given by $f_{n+1}(x) = u(x)$ is a strictly increasing bijection satisfying (2.27).

Now, let us define $h : [x_{-n}, x_{-n+1}] \to \mathbb{R}$, via $h(x) = \frac{a}{b}x - \frac{1}{b}f_{-n}(x)$. It is seen that $h(x_{-n}) = x_{-n+1}$ and $h(x_{-n+1}) = x_{-n}$. Let us take $x > y$ in $[x_{-n}, x_{-n+1}]$. Using the hypothesis (2.25) and the fact that $h(x) - h(y) = -\frac{1}{b}\left(f_{-n}(x) - f_{-n}(y) - a(x-y)\right)$ we obtain

$$\frac{1}{r_2}(x-y) \le h(x) - h(y) \le \frac{1}{r_1}(x-y). \tag{2.28}$$

At this moment we proved that the function $v : [x_{-n}, x_{-n+1}] \to [x_{-n-1}, x_{-n}]$, given via $v(x) = h(x)$ is a strictly increasing bijection. We obtained the strictly increasing bijection $f_{-n-1} = v^{-1}$ namely $f_{-n-1} : [x_{-n-1}, x_{-n}] \to [x_{-n}, x_{-n+1}]$. Inequality (2.28) can be written $\frac{1}{r_2}(x-y) \le f_{-n-1}^{-1}(x) - f_{-n-1}^{-1}(y) \le \frac{1}{r_1}(x-y)$. Taking $f_{-n-1}(x)$ instead of x and $f_{-n-1}(y)$ instead of y we get $\frac{1}{r_2}\left(f_{-n-1}(x) - f_{-n-1}(y)\right) \le x - y \le \frac{1}{r_1}\left(f_{-n-1}(x) - f_{-n-1}(y)\right)$, hence, for all $x > y$ in $[x_{-n-1}, x_{-n}]$

$$r_2(x-y) \le f_{-n-1}(x) - f_{-n-1}(y) \le r_2(x-y). \tag{2.29}$$

At this moment we proved that the function $f_{-n-1} : [x_{-n-1}, x_{-n}] \to [x_{-n}, x_{-n+1}]$, is a strictly increasing bijection which satisfies (2.29). The induction is complete. We succeeded in defining $F_1 : (0, \infty) \to \mathbb{R}$ as in the enunciation. It is clear that F_1 is continuous because $f_n(x_{n+1}) = f_{n+1}(x_{n+1})$ for all $n \in \mathbb{Z}$.

A similar construction furnishes the strictly increasing continuous function $F_2 : (-\infty, 0) \to \mathbb{R}$. Because $\lim_n x_{-n} = 0$ (both right and left limits), it follows that $\lim_{x\to 0} F_1(x) = \lim_{x\to 0} F_2(x) = 0$. At this moment we constructed the continuous and strictly increasing function $f : \mathbb{R} \to \mathbb{R}$ as in enunciation.

All it remains to be proved is the fact that f is a solution, i.e. to be proved that for any $x \in \mathbb{R}$ one has

$$f(f(x)) - af(x) + bx = 0 \tag{2.30}$$

which is obvious for $x = 0$. We shall show (2.30) for $x > 0$ (the proof for $x < 0$ being similar). If $x \geq x_0$, there exists a natural n such that $x \in [x_n, x_{n+1}]$, hence $f(x) \in [x_{n+1}, x_{n+2}]$ and (2.30) becomes

$$f_{n+1}(f_n(x)) - af_n(x) + bx = 0. \tag{2.31}$$

Writing $y = f_n(x)$, (2.31) becomes $f_{n+1}(y) - ay + bf_n^{-1}(y) = 0$, obvious. If $0 < x < x_0$, there exists a natural n such that $x \in [x_{-n-1}, x_{-n}]$ and(2.30) becomes

$$f_{-n}(f_{-n-1}(x)) - af_{-n-1}(x) + bx = 0. \tag{2.32}$$

Let $y \in [x_{-n}, x_{-n+1}]$ such that $x = f_{-n-1}^{-1}(y)$. Then (2.32) becomes $f_{-n}(y) - ay + bf_{-n-1}^{-1}(y) = 0$, obvious.

B. The final part of the proof consists in showing that all solutions are of the form described in part A.

Let $f : \mathbb{R} \to \mathbb{R}$ be a solution. Then f must be strictly increasing and $f(0) = 0$ (Lemma 2.2). We shall prove that $f(x)$ must be of the form $F_1(x)$ for $x > 0$ and $f(x)$ must be of the form $F_2(x)$ for $x < 0$ (see the enunciation). The proof will be given only for $x > 0$, the proof for $x < 0$ being similar.

We take $x_0 > 0$ arbitrarily and write $x_1 = f(x_0)$. If follows that $x_1 \in [r_1x_0, r_2x_0]$, from Lemma 2.2. Now, we construct the sequences $(x_n)_n$ and $(x_{-n})_n$ defined via $x_{n+1} = f(x_n)$ and $x_{-n-1} = f^{-1}(x_{-n})$ with common starting term x_0.

Lemma 2.5 says that $x_n \uparrow \infty$ (strictly) and $x_{-n} \downarrow 0$ strictly, hence $\bigcup_{n\in\mathbb{Z}} [x_n, x_{n+1}] = (0, \infty)$. For any $n \in \mathbb{Z}$, let $F_n : [x_n, x_{n+1}] \to \mathbb{R}$ be the restriction of f to $[x_n, x_{n+1}]$. It follows that $F_n([x_n, x_{n+1}]) = [x_{n+1}, x_{n+2}]$.

Hence $f_n : [x_n, x_{n+1}] \to [x_{n+1}, x_{n+2}]$, given via $f_n(x) = F_n(x)$ is a strictly increasing bijection. The Calibration Theorem 1.4. says that, for all $x > y$ in $[x_0, x_1]$ one has $r_1(x-y) \leq f_0(x) - f_0(y) \leq r_2(x-y)$. It remains to prove (2.24).

First, let us take $x \in [x_{n+1}, x_{n+2}]$ and let $y \in [x_n, x_{n+1}]$ be such that $x = f_n(y) = f(y)$. Because $f(f(y)) - af(y) + by = 0$, we have

$$f_{n+1}(f_n(y)) - af_n(y) + by = 0 \tag{2.33}$$

and (2.33) becomes

$$f_{n+1}(x) - ax + bf_n^{-1}(x) = 0. \tag{2.33'}$$

Finally, let us take $x \in [x_{-n}, x_{-n+1}]$ and let $y \in [x_{-n-1}, x_{-n}]$ such that $x = f_{-n-1}(y) = f(y)$. Because $f(f(y)) - af(y) + by = 0$, we have

$$f_{-n}(f_{-n-1}(y)) - af_{-n-1}(y) + by = 0 \tag{2.34}$$

and (2.34) becomes

$$f_{-n}(x) - ax + bf_{-n-1}^{-1}(x) = 0. \tag{2.34'}$$

$\square$

Remark. Taking $f_0 : [x_0, x_1] \to [x_1, x_2]$, $f_0(x) = r_i x$ (for $x_0 > 0$) and $f_0 : [x_1, x_0] \to [x_2, x_1]$, $f_0(x) = r_j x$ (for $x_0 < 0$), we obtain the solution $f : \mathbb{R} \to \mathbb{R}$ given via

$$f(x) = \begin{cases} r_i x, & \text{if } x \geq 0 \\ r_j x & \text{if } x < 0, \end{cases}$$

where $i, j \in \{1, 2\}$.

Theorem 2.10 (Case $r_2 < r_1 < -1 \Leftrightarrow -1 < r_2 < r_1 < 0$). *All the solutions are obtained as follows:*

We start with an arbitrary $x_0 > 0$ and we choose $x_1 \in [r_2 x_0, r_1 x_0]$. We apply Lemma 2.4 and construct the sequences $(x_n)_n$ and $(x_{-n})_n$. Let $f_0 : [x_0, x_2] \to [x_3, x_1]$ be a strictly decreasing bijection having the property

$$-r_1(x - y) \leq f_0(y) - f_0(x) \leq -r_2(x - y) \tag{C}$$

for all $x > y$ in $[x_0, x_2]$.

We can construct the following strictly decreasing bijections (for any natural n):

$$f_{2n} : [x_{2n}, x_{2n+2}] \to [x_{2n+3}, x_{2n+1}], \ f_{2n}(x) = -ax - bf_{2n-1}^{-1}(x) \tag{2.35}$$

$$f_{2n+1} : [x_{2n+3}, x_{2n+1}] \to [x_{2n+2}, x_{2n+4}], \ f_{2n+1}(x) = -ax - bf_{2n}^{-1}(x) \tag{2.36}$$

$$\begin{gathered} f_{-2n} : [x_{-2n}, x_{-2n+2}] \to [x_{-2n+3}, x_{-2n+1}], \\ f_{-2n}^{-1}(x) = -\tfrac{a}{b}x - \tfrac{1}{b}f_{-2n+1}(x) \end{gathered} \tag{2.37}$$

$$\begin{gathered} f_{-2n-1} : [x_{-2n+1}, x_{-2n-1}] \to [x_{-2n}, x_{-2n+2}], \\ f_{-2n-1}^{-1}(x) = -\tfrac{a}{b}x - \tfrac{1}{b}f_{-2n}(x). \end{gathered} \tag{2.38}$$

Because the reunion of all above mentioned intervals is equal to $\mathbb{R} \setminus \{0\}$, *we can construct* $f : \mathbb{R} \to \mathbb{R}$, *given via*

$$f(x) = \begin{cases} 0, & \text{if } x = 0 \\ f_n(x), & \text{if } x \neq 0, \end{cases}$$

where $0 \neq x$ *belongs to one of the above mentioned intervals which is the domain of definition for* f_n, $n \in \mathbb{Z}$ *(the values at the common endpoints coincide). Then* f *is a solution and all the solutions can be obtained in this way.*

Proof. We begin with the remark that functions f_0 with the stipulated properties do exist. For instance, one can take $f_0(x) = mx + n$, with $f_0(x_0) = x_1$ and $f_0(x_2) = x_3$. We can prove that (see (C))

$$r_1 \geq m = \frac{x_3 - x_1}{x_2 - x_0} \geq r_2$$

using $r_2 x_0 \leq x_1 \leq r_1 x_0$ and expressing x_3 via x_2 and next via x_1.

A. In order to follow more easily the construction we give the start of it, namely

$f_0 : [x_0, x_2] \to [x_3, x_1]$
$f_1 : [x_3, x_1] \to [x_2, x_4]$, $f_1(x) = -ax - bf_0^{-1}(x)$
$f_2 : [x_2, x_4] \to [x_5, x_3]$, $f_2(x) = -ax - bf_1^{-1}(x)$
. . . .
$f_{-1} : [x_1, x_{-1}] \to [x_0, x_2]$, $f_{-1}(x) = -\frac{a}{b}x - \frac{1}{b}f_0(x)$
$f_{-2} : [x_{-2}, x_0] \to [x_1, x_{-1}]$, $f_{-2}(x) = -\frac{a}{b}x - \frac{1}{b}f_{-1}(x)$
$f_{-3} : [x_{-1}, x_{-3}] \to [x_{-2}, x_0]$, $f_{-3}(x) = -\frac{a}{b}x - f_{-2}(x)$
. . . .

We shall proceed using mathematical induction separately for f_n and for f_{-n}.

First part of the induction. We show that, for every natural n, we have the strictly decreasing bijections (defined as above)

$$f_{2n} : [x_{2n}, x_{2n+2}] \to [x_{2n+3}, x_{2n+1}], \text{ and}$$
$$f_{2n+1} : [x_{2n+3}, x_{2n+1}) \to [x_{2n+2}, x_{2n+4}]$$

satisfying for $x > y$ in their domains of definition

$$-r_1(x - y) \leq f_k(y) - f_k(x) \leq -r_2(x - y) \tag{2.39}$$

$$f_k(x_p) = x_{p+1} \tag{2.40}$$

(for x_p = extremity of the domain of definition).

Practically, relation (2.40) proves the continuity of f on $(-\infty, x_1] \cup [x_0, \infty)$.

The first step of the induction. We prove that $f_0 : [x_0, x_2] \to [x_3, x_1]$ and $f_1 : [x_3, x_1] \to [x_2, x_4]$ are strictly decreasing bijections satisfying (2.39) (and, of course (2.40)), For f_1 we have the enunciation. As regards f_1, let $x > y$ in $[x_3, x_1]$ and define $F_1 : [x_3, x_1] \to \mathbb{R}$ via $F_1(x) = -ax - f_0^{-1}(x)$:

$$F_1(y) - F_1(x) = -a(y - x) - b\left(f_0^{-1}(y) - f_0^{-1}(x)\right). \tag{2.41}$$

According to the enunciation one has $-r_1\left(f_0^{-1}(y) - f_0^{-1}(x)\right) \le x - y \le -r_2\left(f_0^{-1}(y) - f_0^{-1}(x)\right)$, which means

$$\frac{x-y}{-r_2} \le f_0^{-1}(y) - f_0^{-1}(x) \le \frac{x-y}{-r_1}. \tag{2.42}$$

In view of $a = -(r_1 + r_2)$, $b = r_1 r_2$, (2.41) becomes $F_1(y) - F_1(x) = -(r_1 + r_2)(x - y) - r_1 r_2 (f_0^{-1}(y) - f_0^{-1}(x))$ and (2.42) gives $F_1(y) - F_1(x) = -(r_1 + r_2)(x - y) + r_1(x - y) = -r_2(x - y)$ and $-(r_1 + r_2)(x - y) + r_2(x - y) = -r_1(x - y) \le F_1(y) - F_1(x)$.

The proof of the first step is finished noticing that

$$F_1(x_3) = -ax_3 - bf_0^{-1}(x_3) = -ax_3 - bx_2 = x_4$$
$$F_1(x_1) = -ax_1 - bf_0^{-1}(x_1) = -ax_1 - bx_0 = x_2$$

(we can define $f_1 : [x_3, x_1] \to [x_2, x_4]$, via $f_1(x) = F_1(x)$).

The second step of the induction. We shall accept that $f_{2n-2} : [x_{2n-1}, x_{2n}] \to [x_{2n+1}, x_{2n-1}]$ and $f_{2n-1} : [x_{2n+1}, x_{2n-1}] \to [x_{2n}, x_{2n+2}]$ are strictly decreasing bijections as in the enunciation satisfying (2.39) and (2.40). We must prove that $f_{2n} : [x_{2n}, x_{2n+2}] \to [x_{2n+3}, x_{2n+1}]$ and $f_{2n+1} : [x_{2n+3}, x_{2n+1}] \to [x_{2n+2}, x_{2n+4}]$ are strictly decreasing bijections satisfying (2.39) and (2.40).

We consider the function $u : [x_{2n}, x_{2n+1}] \to \mathbb{R}$ given via $u(x) = -ax - f_{2n-1}^{-1}(x)$ and notice that

$$u\left(x_{2n}\right) = -ax_{2n} - bx_{2n-1} = x_{2n+1}$$
$$u\left(x_{2n+2}\right) = ax_{2n+2} - bx_{2n+1} = x_{2n+3}.$$

As the same time, taking in to account (2.39) for $k = 2n-1$, the same procedure with that one used in the proof of Theorem 2.9 furnishes $-r_1(x-y) \leq u(y) - y(x) \leq -r_2(x-y)$. The last double inequality shows that the function $f_{2n} : [x_{2n}, x_{2n+1}] \to [x_{2n+3}, x_{2n+1}]$ given via $f_{2n}(x) = u(x)$ is a strictly decreasing bijection satisfying (2.39) and (2.40).

We shall use this fact further for the properties of f_{2n+1}. Namely, we define $v : [x_{2n+3}, x_{2n+1}] \to [x_{2n+2}, x_{2n+4}]$, given via $v(x) = -ax - bf_{2n}^{-1}(x)$ and notice that $v(x_{2n+3}) = x_{2n+4}$, $v(x_{2n+1}) = x_{2n+2}$ and (with same procedure) $-r_1(x-y) = v(y) - v(x) \leq -r_2(x-y)$

At this moment we proved that the function $f_{2n+1} : [x_{2n+3}, x_{2n+1}] \to [x_{2n+2}, x_{2n+4}]$, given via $f_{n+1}(x) = v(x)$ is a strictly decreasing bijection satisfying (2.39) and (2.40). The announced induction is complete.

Second part of the induction (concerning f_{-n}). We show that, for every natural n, we have the strictly decreasing bijections defined as above $f_{-2n} : [x_{-2n}, x_{-2n+2}] \to [x_{-2n+3}, x_{-2n+1}]$ and $f_{-2n-1} : [x_{-2n+1}, x_{-2n-1}] \to [x_{-2n}, x_{-2n+2}]$ satisfying (2.39) and (2.40). Notice that (2.40) will prove also the continuity of f on $[x_1, 0) \cup (0, x_0]$. Because $x_{-n} \to 0$ (both from the right and from the left) it will follow that the function f defined in the enunciation is continuous.

The first step of the induction. We prove that $f_0 : [x_0, x_2] \to [x_3, x_1]$ and $f_{-1} : [x_1, x_{-1}] \to [x_0, x_2]$ are strictly decreasing bijections satisfying (2.39).

To this aim, we define $h : [x_0, x_2] \to \mathbb{R}$ via $h(x) = -\frac{a}{b}x - \frac{1}{b}f_0(x)$. For $x > y$ in $[x_0, x_2]$ one has $h(y) - h(x) = \frac{1}{b}\left(a(x-y) + f_0(x) - f_0(y)\right)$. Because $r_2(x-y) \leq f_0(x) - f_0(y) \leq r_1(x-y)$ we get $\frac{1}{b}\left(a(x-y) + r_2(x-y)\right) \leq h(y) - h(x) \leq \frac{1}{b}\left(a(x-y) + r_1(x-y)\right)$.

In view of $a = -(r_1 + r_2)$, $b = r_1 r_2$, we get

$$\frac{x-y}{-r_2} \leq h(y) - h(x) \leq \frac{x-y}{-r_1}, \tag{2.43}$$

which shows the fact that h is strictly decreasing.

We notice that

$$h(x_0) = -\frac{a}{b}x_0 - \frac{1}{b}f_0(x_0) = -\frac{a}{b}x_0 - \frac{1}{b}x_1 = x_{-1}$$
$$h(x_2) = -\frac{a}{b}x_2 - \frac{1}{b}f_0(x_2) = -\frac{a}{b}x_2 - \frac{1}{b}x_3 = x_1.$$

Now, we have the bijection $H : [x_0, x_2] \to [x_1, x_{-1}]$, $H(x) = -\frac{a}{b}x - \frac{1}{b}f_0(x)$ and it is possible to define the strictly decreasing bijection $f_{-1} : [x_1, x_{-1}] \to [x_0, x_2]$ via $f_{-1} = H^{-1}$. Taking in (2.43): $x = f_{-1}(x')$, $y = f_{-1}(y')$ we have $x' < y'$ in $[x_1, x_{-1}]$ and we get

$$\frac{f_{-1}(x') - f_{-1}(y')}{-r_2} \leq y' - x' \leq \frac{f_{-1}(x') - f_{-1}(y')}{-r_1}$$

which shows that $-r_1(y' - x') \leq f_{-1}(x') - f_{-1}(y') \leq -r_2(y' - x')$ and the first step of the induction is complete.

The second step of the induction. We shall accept that $f_{-2n+2} : [x_{-2n+2}, x_{-2n+4}] \to [x_{-2n+5}, x_{-2n+3}]$ and $f_{-2n+1} : [x_{-2n+3}, x_{-2n+1}] \to [x_{-2n+2}, x_{-2n+4}]$ are strictly decreasing bijections as in the enunciation satisfying (2.39) and (2.40). We must prove that $f_{-2n} : [x_{-2n}, x_{-2n+2}] \to [x_{-2n+3}, x_{-2n+1}]$ and $f_{-2n-1} : [x_{-2n+1}, x_{-2n-1}] \to [x_{-2n}, x_{-2n+2}]$ are strictly decreasing bijections satisfying (2.39) and (2.40).

The proof is similar to the proof for f_{-n} of Theorem 2.9. One uses the properties of f_{-2n+1} and one constructs the function $H_{-2n} : [x_{-2n+3}, x_{-2n+1}] \to \mathbb{R}$ given via: $H_{-2n}(x) = -\frac{a}{b}x - \frac{1}{b}f_{-2n+1}(x)$. It is seen that $H_{-2n}(x_{-2n+3}) = x_{-2n+2}$ and $H(x_{-2n+1}) = x_{-2n}$. Using (2.39) for f_{-2n+1} one gets (proof is similar to the proof of theorem 2.9 and the proof of the first step of induction)

$$\frac{x - y}{-r_2} \leq H_{-2n}(y) - H_{-2n}(x) \leq \frac{x - y}{-r_1} \tag{2.44}$$

for $x > y$ in $[x_{-2n+3}, x_{-2n+1}]$.

This implies that the function $h_{-2n} : [x_{-2n+3}, x_{-2n+1}] \to [x_{-2n}, x_{-2n+2}]$ is a strictly decreasing bijection, where h_{-2n} is given via $h_{-2n}(x) = H_{-2n}(x)$. We can define $f_{-2n} = h_{-2n}^{-1}$ and using (2.44) we get $-r_2(x - y) \leq f_{-2n}(y) - f_{-2n}(x) \leq -r_1(x - y)$.

Using the properties of f_{-2n}, one constructs f_{-2n-1} and one checks its suitable properties. The induction is complete. At this moment, the construction of f is complete. To check the fact that f is a solution, the computations are exactly those from the proof of Theorem 2.9 because, in order to write $f \circ f(x)$ one can write $f_{n+1} \circ f_n$ for some $n \in \mathbb{Z}$.

B. The final part of the proof consists in proving that all solutions are of the form described in part **A.**

Let f be a solution. Then f must be strictly decreasing and $f(0) = 0$ (Lemma 2.3).

We pick arbitrarily $x_0 > 0$. Let us construct (see Lemma 2.4) the sequences $(x_n)_n$, $(x_{-n})_{-n}$ as follows $x_{n+1} = f(x_n)$ and $x_{-n-1} = f^{-1}(x_{-n})$ with common starting point x_0. Lemma 2.3 tells that $x_1 \in [r_2 x_0, r_1 x_0]$. The other properties of these sequences are given in Lemma 2.4. For any natural n we can construct the following functions:

a) $F_{2n} : [x_{2n}, x_{2n+2}] \to \mathbb{R}$, $F_{2n}(x) = f(x)$,

b) $F_{2n+1} : [x_{2n+3}, x_{2n+1}] \to \mathbb{R}$, $F_{2n+1}(x) = f(x)$,

c) $F_{-2n} : [x_{-2n}, x_{-2n+2}] \to \mathbb{R}$, $F_{-2n}(x) = f(x)$,

d) $F_{-2n-1} : [x_{-2n+1}, x_{-2n-1}] \to \mathbb{R}$, $F_{-2n-1}(x) = f(x)$.

The functions defined above are strictly decreasing. Due to the fact that $F_{2n}(x_{2n}) = f(x_{2n}) = x_{2n+1}$ and $F_{2n}(x_{2n+2}) = f(x_{2n+2}) = x_{2n+3}$ we can define the strictly decreasing bijection $f_{2n} : [x_{2n}, x_{2n+2}] \to [x_{2n+3}, x_{2n+1}]$ given via $f_{2n}(x) = F_{2n}(x)$.

In the same way, we define the strictly decreasing bijections f_{2n+1}, f_{-2n}, f_{-2n-1}.

Because of the Calibration Theorem 1.4 we have (C).

All it remains to be proved consists in the relations (2.35), (2.36), (2.37), (2.38). Studying carefully the domains of definition for our functions (practically the orbits $f_0(x)$, $f_1(x)$, $f_2(x)$, $\ldots$, $f_n(x)$, $\ldots$ and $f_{-1}^{-1}(x)$, $f_{-2}^{-1}(x)$, $f_{-3}^{-1}(x)$, $\ldots$, $f_{-n}^{-1}(x)$, $\ldots$) we can see that the relations (2.33) and (2.34) remain true and they imply relations (2.33′) and (2.34′) which are exactly (2.35), (2.36), (2.37), (2.38). $\square$

Remark. Taking $f_0 : [x_0, x_2] \to [x_3, x_1]$, $f_0(x) = r_i x$, we obtain the solutions $S(r_i)$, $i = 1, 2$.

Lemma 2.11. *In case $0 < r_1 < 1 < r_2$, assume that a solution f has the property that $f(x_0) = r_1 x_0$ or $f(x_0) = r_2 x_0$, for some $x_0 \in \mathbb{R}$. Then $f(0) = 0$.*

Proof. a) Assume first the existence of $x_0 \in \mathbb{R}$ such that $f(x_0) = r_1 x_0$. It follows that $f(r_1 x_0) = (r_1 + r_2) r_1 x_0 - r_1 r_2 x_0 = r_1^2 x_0$. We can continue:

$$f\left(r_1^2 x_0\right) = f\left(f\left(r_1 x_0\right)\right) = (r_1 + r_2) r_1^2 x_0 - r_1 r_2 r_1^2 x_0 = r_1^3 x_0$$

and, inductively one can prove that for any natural n one has $f\left(r_1^n x_0\right) = r_1^{n+1} x_0$. Because f is continuous, we pass to n-limit and obtain $f(0) = 0$.

b) Now, assume that $f(x_0) = r_2 x_0$ for some $x_0 \in \mathbb{R}$. This implies that $f\left(f\left(f^{-1}(x_0)\right)\right) = (r_1+r_2)f\left(f^{-1}(x_0)\right) - r_1 r_2 f^{-1}(x_0)$ which means $r_2 x_0 = f(x_0) = (r_1+r_2)x_0 - r_1 r_2 f^{-1}(x_0) \Rightarrow f^{-1}(x_0) = \frac{x_0}{r_2}$. Because for f^{-1} the roots of the characteristic equations are $0 < \frac{1}{r_2} < 1 < \frac{1}{r_1}$, we can apply the results of part a) and obtain that

$$f^{-1}\left(\frac{x_0}{r_2^n}\right) = \frac{x_0}{r_2^{n+1}}.$$

Passing to $n-$limit we get $f^{-1}(0) = 0$ which implies $0 = f(0)$. □

Theorem 2.12. *Assume $0 < r_1 < 1 < r_2$ and let f be a solution with the property $f(0) \neq 0$. Then either $f(x) > x$ for any $x \in \mathbb{R}$ or $f(x) < x$ for any $x \in \mathbb{R}$.*

I. Assume that $f(x) > x$ for all $x \in \mathbb{R}$. Then f can be obtained as follows:

We construct the sequences $(x_n)_n$ and $(x_{-n})_n$ according to Lemma 2.6, where we take $x_0 = 0$ and $x_1 > 0$ arbitrary (the conditions of Lemma 2.6 are fulfilled). We consider a strictly increasing bijection $f_0 : [0, x_1] \to [x_1, x_2]$ such that $r_1(x-y) \leq f_0(x) - f_0(y) \leq r_2(x-y)$ for all $x > y$ in $[0, x_1]$. Then, for any natural n one can construct the bijections $f_n : [x_n, x_{n+1}] \to [x_{n+1}, x_{n+2}]$ and $f_{-n} : [x_{-n}, x_{-n+1}] \to [x_{-n+1}, x_{-n+2}]$ defined via $f_{n+1}(x) = ax - bf_n^{-1}(x)$ and $f_{-n-1}^{-1}(x) = \frac{a}{b}x - \frac{1}{b}f_{-n}(x)$.

Finally, for any $x \in \mathbb{R} = \bigcup_{n\in\mathbb{Z}} [x_n, x_{n+1}]$, we have, for some natural n:

– either $x \in [x_n, x_{n+1}]$ and $f(x) = f_n(x)$

– or $x \in [x_{-n}, x_{-n+1}]$ and $f(x) = f_{-n}(x)$.

II. Assume that $f(x) < x$ for any $x \in \mathbb{R}$. Then $f^{-1}(x) > x$ for any $x \in \mathbb{R}$ and f^{-1} can be constructed according to part I.

Proof. We prove I. It is seen that f cannot have fixed points. Indeed, if l would be a fixed point, we should have (writing with $a > 0$ the fundamental equation) $(f \circ f)(l) - af(l) + bl = l(1-a+b) = 0$. Because $l \neq 0$, it would follow that $1 - a + b = 0$ which is not true, because $r_1 \neq 1$, $r_2 \neq 1$.

We have seen (see the proof Theorem 2.9) that functions f_0 with the stipulated properties do exist.

The remaining part of the proof is exactly like the proof of Theorem 2.9 construction of F_1 (here we have $(-\infty, \infty)$ instead of $(0, \infty)$.

Point II is obvious.

Now, let us prove that all solutions f having the property that $f(x) > x$ for all $x \in \mathbb{R}$ can be constructed like in the enunciation. Let f be such a solution. It is clear that f is strictly increasing. Writing $x_{n+1} = f(x_n)$ and $x_{-n-1} = f^{-1}(x_{-n})$, we define the sequences $(x_n)_n$ and $(x_{-n})_n$ such that $x_n \uparrow \infty$ (strictly) and $x_{-n} \downarrow -\infty$ (strictly), according to Lemma 2.6. More precisely, we have $\bigcup_{n=0}^{\infty} [x_n, x_{n+1}] = [0, \infty)$ and $\bigcup_{n=1}^{\infty} [x_{-n}, x_{-n+1}] = (-\infty, 0]$.

The remaining part of this proof is exactly like in the proof of Theorem 2.9. $\square$

Theorem 2.13. *Assume $0 < r_1 < 1 < r_2$ and let f be a solution with the property $f(0) = 0$. Then f must be one of the following four functions: $S(r_1)$, $S(r_2)$, S_{12} and S_{21}. Here $S_{12}, S_{21} : \mathbb{R} \to \mathbb{R}$ are given via*

$$S_{12}(x) = \begin{cases} r_1 x, & \text{if } x \le 0 \\ r_2 x, & \text{if } x > 0 \end{cases} \quad \text{and} \quad S_{21}(x) = \begin{cases} r_2 x, & \text{if } x \le 0 \\ r_1 x, & \text{if } x > 0. \end{cases}$$

Proof. The proof will follow the proof of Theorem 2.7. Using the fact that $f(0) = 0$, we introduce the same functions f and h, but now $g \uparrow$ and $h \downarrow$. The relations from (2.9) remain valid.

For $x_0 \neq 0$, writing again $x_{2n} = f^{2n}(x_0)$ one can prove that if $x_0 > 0$ one has $r_1^{2n} x_0 \le x_{2n} \le r_2^{2n} x_0$ and for $x_0 < 0$ one has $r_2^{2n} x_0 \le x_{2n} \le r_1^{2n} x_0$. For $x_0 > 0$ one has again either $x_{2n} \uparrow \infty$ or $x_{2n} \downarrow 0$ strictly and for $x_n < 0$ one has either $x_{2n} \uparrow 0$ or $x_{2n} \downarrow -\infty$ strictly, using (2.10) which remains valid.

The equivalences (2.11) and (2.12) remain valid and also equalities (2.13) and (2.14). We arrive at the same conclusion, namely that for any $x_0 \in \mathbb{R}$ one must have either $f(x_0) = r_1 x_0$ or $f(x_0) = r_2 x_0$.

Let f be a solution which is not $S(r_1)$ or $S(r_2)$. There exist two points $x_1 \neq x_2$ such that $f(x_1) = r_1 x_1$ and $f(x_2) = r_2 x_2$. Firstly we shall assume $x_1 < x_2$. We introduce the sets: $A_1 = \{x \in \mathbb{R} \mid f(x) = r_1 x\}$ and $A_2 = \{x \in \mathbb{R} \mid f(x) = r_2 x\}$ which are non empty and closed, because f is continuous. We have $A_1 \ni x_1 < x_2 \in A_2$ and, of course, $A_1 \cup A_2 = \mathbb{R}$, $A_1 \cap A_2 \supset \{0\}$. Let $a_1 = \sup A_1$ and $a_2 = \inf A_2$ (hence $x_1 \le a_1 \in A_1$ and $x_2 \ge a_2 \in A_2$). Because $(a_1, \infty) \subset A_2$, we can find two sequences $(u_n)_n$, $(v_n)_n$ such that $u_n \underset{n}{\to} a_1$, $v_n \underset{n}{\to} a_1$ and $u_n \in A_1$, $v_n \in A_2$. Hence $f(u_n) = r_1 u_n \underset{n}{\to} f(a_1)$ and $f(v_n) = r_2 v_n \underset{n}{\to} f(a_1)$. This means that $f(a_1) = r_1 a_1 = r_2 a_1$ and this is possible if and only if $a_1 = 0$.

In the same way one proves that $a_2 = 0$.

At this moment we have proved that $A_1 = (-\infty, 0]$ and $A_2 = [0, \infty)$, hence $f = S_{12}$.

In the same way, if $x_1 > x_2$ we prove that $f = S_{21}$. $\square$

Using Theorem 2.12 and 2.13, we have finally

Theorem 2.14 (Case $0 < r_1 < 1 < r_2$)**.** *The solutions are* $S(r_1)$, $S(r_2)$, S_{12}, S_{21} *and all the solutions given by Theorem* 2.12.

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